

Overview

This manual is intended to:

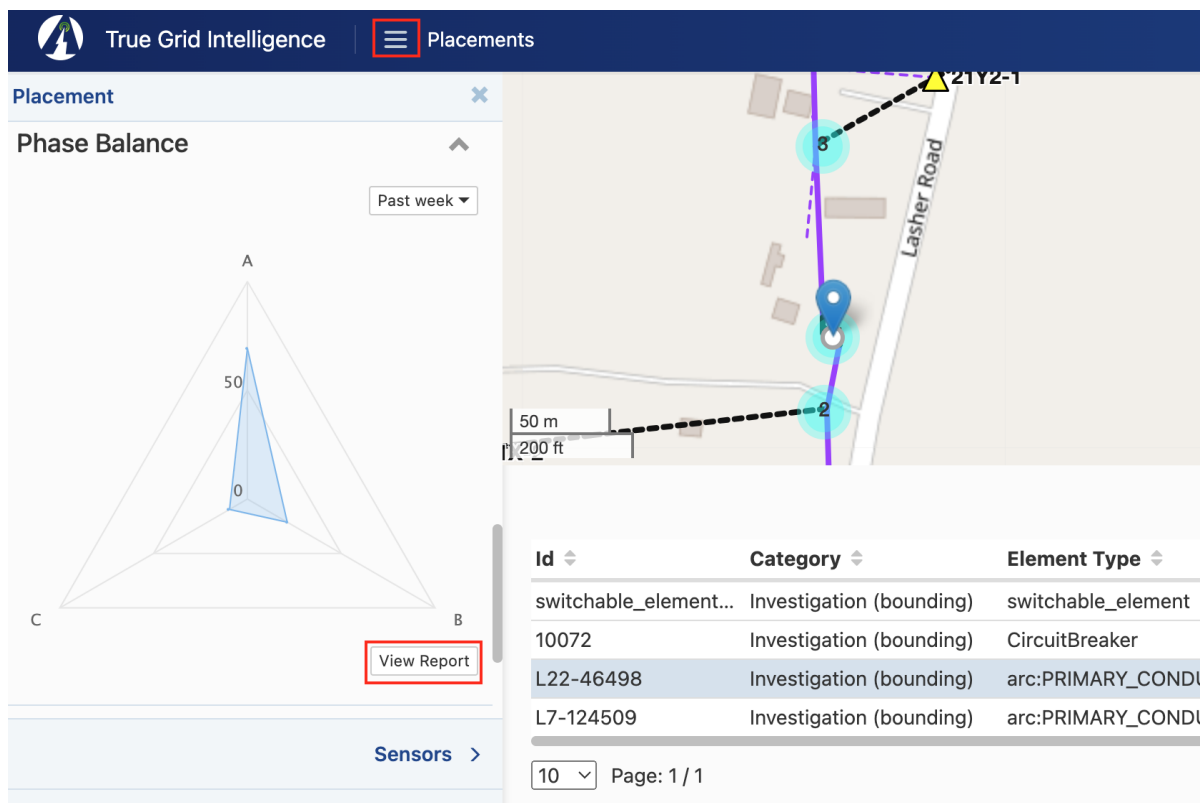
- Provide instructions to create the selected use case in True Grid Intelligence (TGI).

For more details about Power Factor Analysis and how it can be analyzed using Awesense's Energy Transition Platform, please refer to the [UC02-01 - Power Factor Analysis.pdf](#) document.

TGI Use Case Notes

This document shows the implementation process of mentioned use case in the TGI visualization interface along with a description of how to create such a use case on your data.

The Power Factor Analysis can be found directly in the TGI for a Placement if the Placement has per-phase measured data. The easiest way is to find the Placement via the TGI Explorer in the main menu (select from the main menu - top hamburger icon) and then select the Placement from the map or table. Clicking on a Placement will open a sidebar on the left side with additional information, and at the bottom of this sidebar is the "Phase Balance" section where clicking on "View Report" will display this report.



Power Factor Analysis - TGI Use Case Notes

[UC02-01] - Awesense Energy Transition Platform Use Case

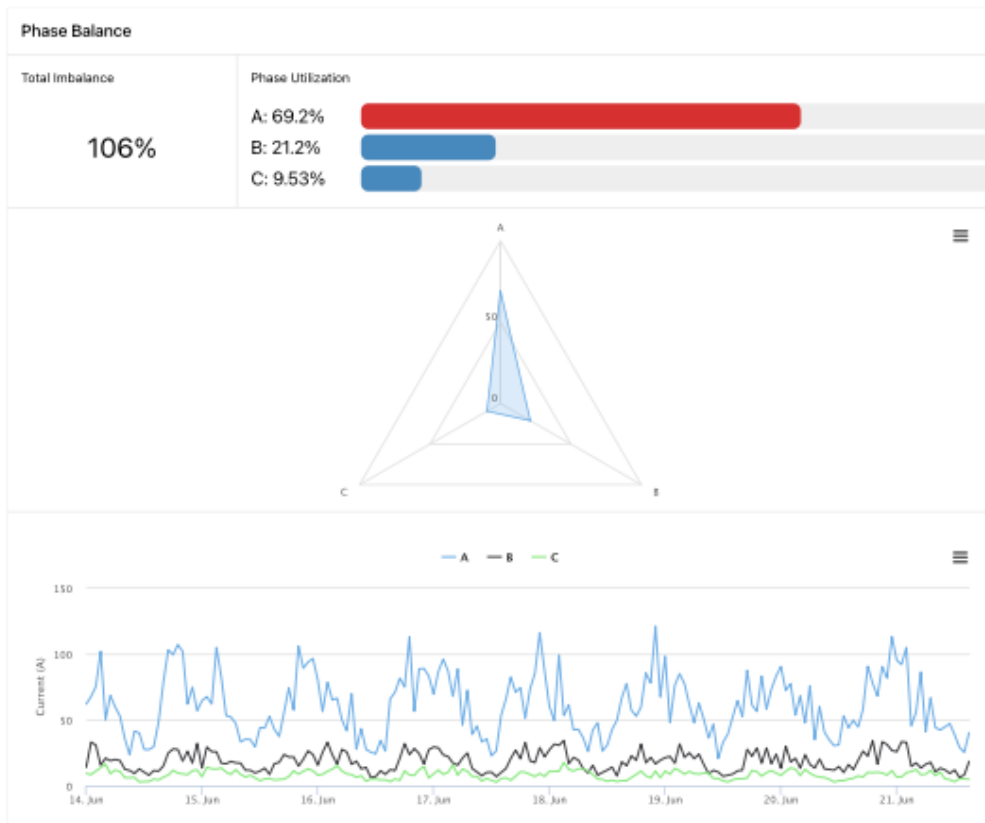


The first part of the report focuses on general information about Placement. The second part focuses on phase balance, which is covered more in another use case notes document. The third part of TGI's report focuses specifically on the power factor.

Placement Technical Analysis

Placement Overview	
Placement type:	Investigation (bounding)
Investigation:	Investigation of L22-46498
Element ID:	L22-46498
Location:	--
Phase A:	MB467L1
Phase B:	MB467L2
Phase C:	MB467L3
is_operation_status:	Yes
label:	L22
make_model:	336 MCM ACSR
materials:	Al
name:	46498
ph_ph_voltage:	14000
rating_a:	529
voltage_level:	MV
wire_diameter:	18.288

14 Jun, 2023 12:00 AM - 21 Jun, 2023 3:08 PM

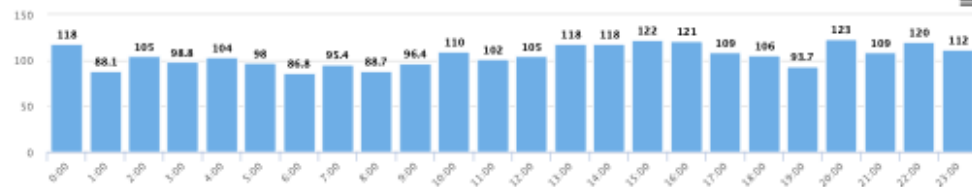


Power Factor Analysis - TGI Use Case Notes

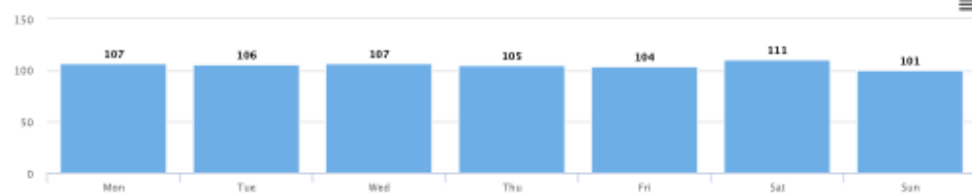
[UC02-01] - Awesense Energy Transition Platform Use Case



Average imbalance grouped by hour



Average imbalance grouped by day of week



Power Factor

Average PF

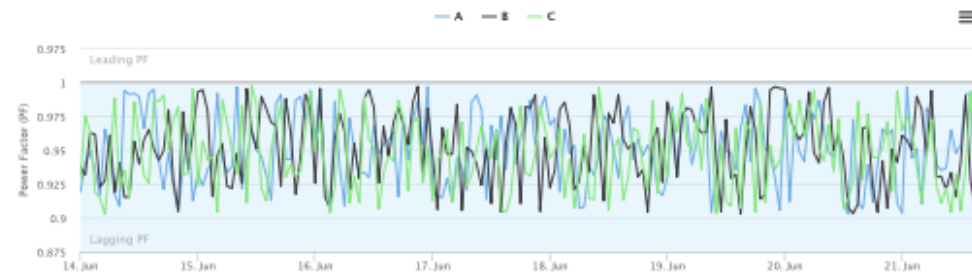
0.95 Lead/Lag

Average PF by phase

A: 0.951

B: 0.952

C: 0.948

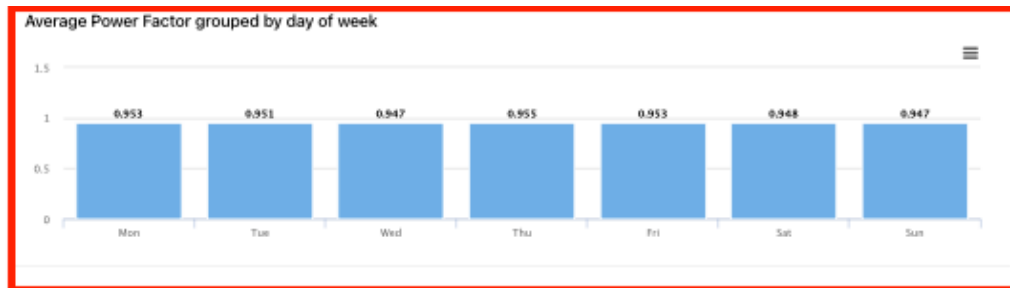


Average Power Factor grouped by hour



Power Factor Analysis - TGI Use Case Notes

[UC02-01] - Awesense Energy Transition Platform Use Case



The basic interface of the report offers the possibility to change the different time periods. Placement's power factor data can be analyzed for two basic charts (histograms):

1. The average for an hour of the day.
2. The average for the day of the week.

These two histograms offer a way of correlating the poor power factor with a specific day or hour. This then can lead to better identification of reactive power consumption and better power factor correction actions.

General Note

This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different. Please contact tools@awesense.com for more information.